# Session 1: Introduction générale au XML et à la TEI

## Atelier Humanités Numériques

## Lyon

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2 Schemata
- Interests of using schemata
- Schemas and the TEI I
- Schemas and the TEI II: ODD's
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# What is the use of a schema?

- A schema checks if the data is correctly formatted.
- A schema makes assumptions about what is legal and what is illegal in an XML data structure
- Applied to some text encoding, a schema is a model of the “text”, that is, a **formal representation** of the text.
- From a pragmatic point of view, it helps harmonizing your encoding and is an **important step towards interoperability**

# Integrating the schema in your encoding sessions

Some XML editors use the schema as a tool for easing the encoding:

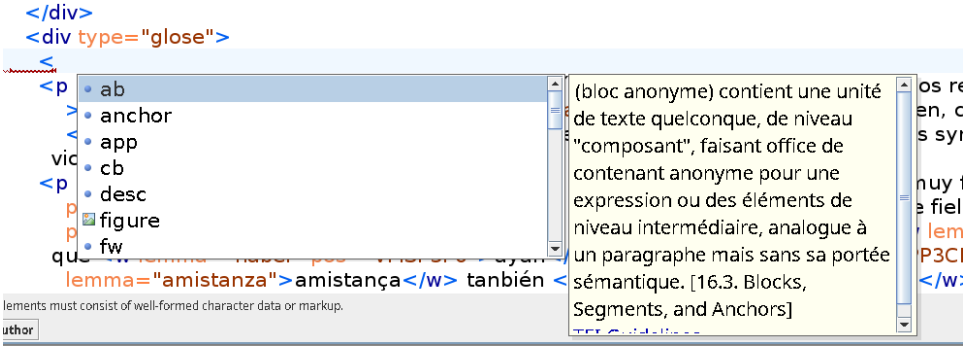


Figure: Oxygen XML Editor allows you to choose between the allowed elements

# Schemas

- A schema is a document that is used to control the quality of some encoding.
- The TEI provides **rules** (the *Guidelines*, human readable), and their **formalization** (*schemas*, machine readable) to check the validity of a given document
- From an abstract point of view, the schema represents the formalization of *your* modelling of a given text or genre.

# Schemas

```
TEI  text  body  div  div  div  lb
1803   <body>
1804     <div type="livre" n="1">
1805       <div type="partie" n="2">
1806         <div type="chapitre" n="1"><pb n="" facs="../input_files/16369405.jpg"/><fw
1807           type="titre_courant"/><lb break="?" rend="rubric"
1808           xml:id="elem_eSc_line_99cc2a93"/>Capitulum . i. lquno diuiduntur potencie<lb
1809           break="yes" rend="rubric"
1810           xml:id="elem_eSc_line_7f955f55"/>anime et in quibus potenciis habet esse uirtutes. <lb
1811           break="yes"
1812           xml:id="elem_eSc_line_6ea8e267"/>ostquam auxiliante deo compleui<lb break="no"
1813           xml:id="elem_eSc_line_df9d9bc9"/>mus primampartem huius primi libri inquo<lb
1814           break="yes"
1815           xml:id="elem_eSc_line_b36db744"/>agitur de regimine sui ostendentes<lb break="no"
1816           xml:id="elem_eSc_line_2c352677"/>i quo reges et p̃ncipes suam felici □□ poner<lb
1817           break="no"
1818           xml:id="elem_eSc_line_54b88267"/>e d□ quia non decet eos suum finem pone in din<lb
1819           break="no"
1820           xml:id="elem_eSc_line_a6468504"/>ciis. nec in ciuili potentia. nec in aliquibus
1821         talibus sed omni<lb break="no"
1822           xml:id="elem_eSc_line_5efa6adc"/>bus hiis ut supra plenius perba debet uti tanquam<lb
1823           break="yes"
1824           xml:id="elem_eSc_line_33a63380"/>organis ad felicitatem sed poneredz in a<lb
1825           break="no"
1826           xml:id="elem_eSc_line_c991f46c"/>ctu pruden □ prout talis actus est inpatu□□<lb
1827           break="yes"
1828           xml:id="elem_eSc_line_388d53ce"/>a cari □□ Nam tunc reges habet felicita<lb
1829           break="no"
1830           xml:id="elem_eSc_line_4b955e3e"/>tem suo statui debitas et condignam quando<lb
1831           break="yes"
1832           xml:id="elem_eSc_line_c422ed80"/>instigante dei dilectione secundum prudenciam<lb
1833           break="yes"
text not allowed here; expected the element end-tag or element "ab", "addSpan", "alt", "altGrp", "anchor", "annotationBlock", "app", "argument", "bibl", "biblFull", "biblStruct", "byline", "camera", "caption", "castList",
```

Figure: This fragment is *well-formed*, but it is not *valid* according to the XML-TEI schema. A paragraph `p` or an anonymous block `ab` should wrap the lines.

# Schema formats

There are multiple schema formats. The most important are:

- Document Type Definition (DTD), already mentionned before
- XML schema (XSD format)
- Relax NG (RNG) schemas and its compact version (RNC)

The former is less used nowadays but can prove usefull for creating shortcut elements thanks to **XML entities** (e.g. for abbreviation encoding)

```
</div>
<div type="chapitre" n="9" xml:id="Mad_A_3_3_9">
 <head n="aPtIUvCUOv">Capitulo .ix°. <w lemma="do" pos="RG">&doblique-d;o</w>
  mue&slong-s;tra &qtrema-qua;<lb facs="#facs_maBaQz" xml:id="elem_maBaQz" break="no"
  />les &e-e; quantas co&                                  e-d;e
    <w><sic>pe&slong-                                      ko41Kz" xml:id="elem_ko41Kz"
   break="yes"/>en&rien                                    "NCFS000">batalla</w>
  quan&doblique-d;o &slo                                   ="#facs_mxdFHc"
   xml:id="elem_mxdFHc                                     </head>
 <div type="traduction">
```

```
<choice n="qtrema-qua" ana="
#normalisation" xmlns="http:
//www.tei-c.org/ns/1.0"
><abbr>q̈</abbr><expan>q<ex>ua
</ex></expan></choice>
```

Figure: A possible way to encode abbreviations using XML entities / DTD definitions

2 Schemata
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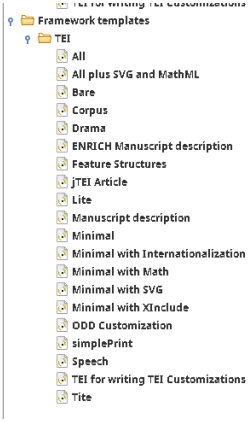
# TEI subsets




- There are multiple TEI schemas available.
- As a scientific standard, the TEI needs to be adapted to the users needs.

# Why should you create your own TEI customization ?

- Consider the TEI as a tool to model your textual object
- You probably won't need all the intellectual tools provided by the TEI Consortium.
- As medievalists, you might want to discard the elements belonging to the "Transcriptions of Speech" section of the TEI (alas!)
- Your schema will be the formalization of your way of seeing the text, and you need it to be as specific as possible.

# Two main methods

- The inclusion method: start from scratch, add the elements you need
- The exclusion method: start from a predefined schema, remove the elements you won't need

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# Two main methods

- The inclusion method: start from scratch, add the elements you need
- The exclusion method: start from a predefined schema, remove the elements you won't need
- I would recommended to adopt the first approach...
- or at least starting from a preset specification that meets some of your interests (we'll see this later).
- Mastering the Guidelines is essential to know what is possible for your project.

# Don't reinvent the wheel !

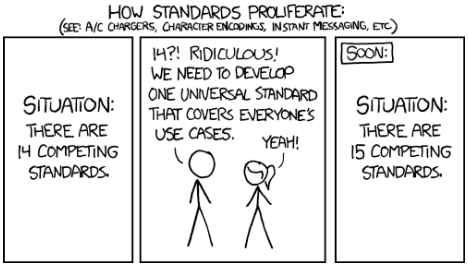


Figure: `https://xkcd.com/927/`

- You have to specify, and adapt the TEI rules to your own project
- However, the modelling problems you will face are probably *not* new
- Please don't reinvent the wheel !
- Search for what's been done for similar problems and try the provided solutions !
- And (for a start), avoid inventing/creating new elements.

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- You want now to explain your rules to your supervisor, to your colleagues and to your children. You need to write this *in natural language*.

# Is that all, folks ?

- Create a specific schema is a good start.
- What's missing then ? *Documentation*.
- You want now to explain your rules to your supervisor, to your colleagues and to your children. You need to write this *in natural language*.
- What could be better than a TEI document to describe another TEI document?

# Here comes the ODD

- ODD stands for "One Document Does it All"
- An ODD is a TEI document that is used to create both the documentation and the schema in a single document.
- Important reading: Lou Burnard. "What Is TEI Conformance, and Why Should You Care?" In: *Journal of the Text Encoding Initiative* 12 (2019). ISSN: 2162-5603

# One Document to Rule Them All

An ODD contains:

- the machine-readable elements used to *validate* an edition
- the human-readable elements used to *document* it

**Text Body**


- 1 The TEI Infrastructure
- 2 The TEI Header
- 3 Elements Available in All TEI Documents
- 4 Default Text Structure
- 5 Characters, Glyphs, and Writing Modes
- 6 Verse
- 7 Performance Texts
- 8 Transcriptions of Speech
- 9 Computer-mediated Communication
- 10 Dictionaries
- 11 Manuscript Description
- 12 Representation of Primary Sources
- 13 Critical Apparatus
- 14 Names, Dates, People, and Places
- 15 Tables, Formulæ, Graphics, and Notated Music
- 16 Language Corpora
- 17 Linking, Segmentation, and Alignment
- 18 Simple Analytic Mechanisms
- 19 Feature Structures
- 20 Graphs, Networks, and Trees
- 21 Non-hierarchical Structures
- 22 Certainty, Precision, and Responsibility
- 23 Documentation Elements
- 24 Using the TEI


- The TEI is organized by *thematic* modules

# TEI as modules

- 22 modules
- Each module is binded to a list of elements. Deactivate a module and those elements won't be accessible anymore.
- 4 main modules: `tei`, `core`, `header`, and `textstructure`, almost mandatory
- The other module you might want to take a look at are: `msdescription`, (`transcr`) and `textcrit`

# An ODD is NOT a schema

- It is a **TEI document** used to produce schemata.
- It needs to be **converted** to a schema (and a documentation when applicable)
- The resulting schema (in general, in RNG) is the document you will use to validate your XML documents.
- Don't try to validate your files against ODD's !

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# Producing ODD's and schemas with ROMA

- ROMA, maintained by Raffaele Viglianti (lead), Peter Stadler and Bryan Wang.
- ROMA is a simple interface for designing ODDs and creating schemas
- URL: `https://roma.tei-c.org/`

# A quick look at our ODD

With ROMA, you can easily design your own ODD:

- Include or exclude specific modules
- Completely remove elements (they won't be allowed anymore)
- Constrain attribute types, or attribute values
- Make some attributes mandatory
- etc.

# Producing ODD's and schemas with ROMA

- I'm interested in the encoding of a medieval manuscript. How can I create the most adapted ODD with ROMA?
- I want to make sur the user uses an attribute `@n` for each page beginning element (`pb`). How can I do ?
- I want to remove the `div1`, `div2`, ..., `div7` elements, so that divisions are marked up with typed and identified `div` elements only.

# Producing ODD's and schemas with ROMA

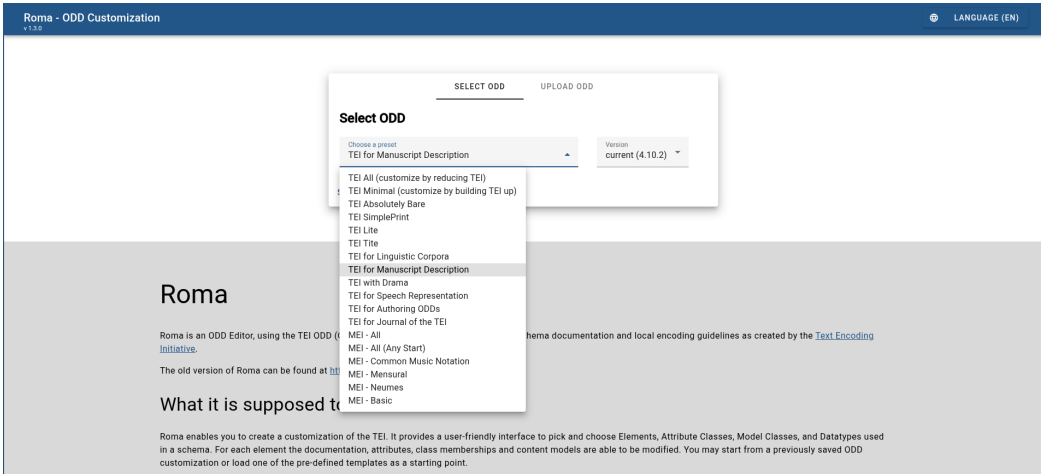


Figure: ROMA main page: presets

# Producing ODD's and schemas with ROMA

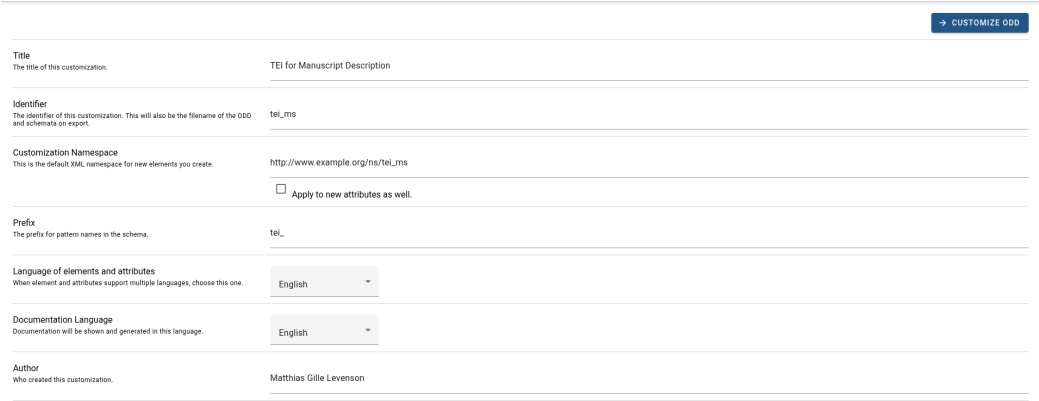


Figure: Configuring your ODD

# Producing ODD's and schemas with ROMA

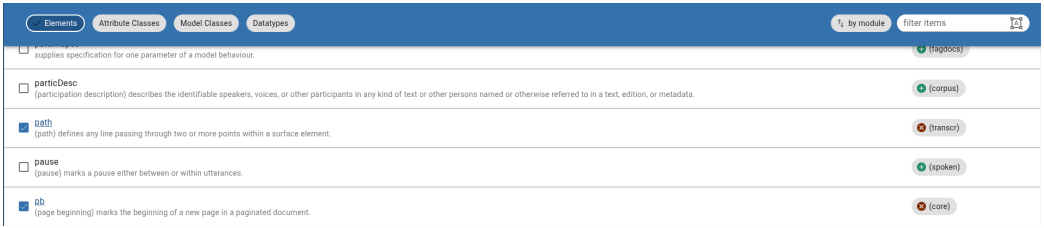


Figure: Find the element you want to modify (here, pb)

# Producing ODD's and schemas with ROMA

Figure: Choose the feature you want to adapt (documentation or attribute)

# Producing ODD's and schemas with ROMA

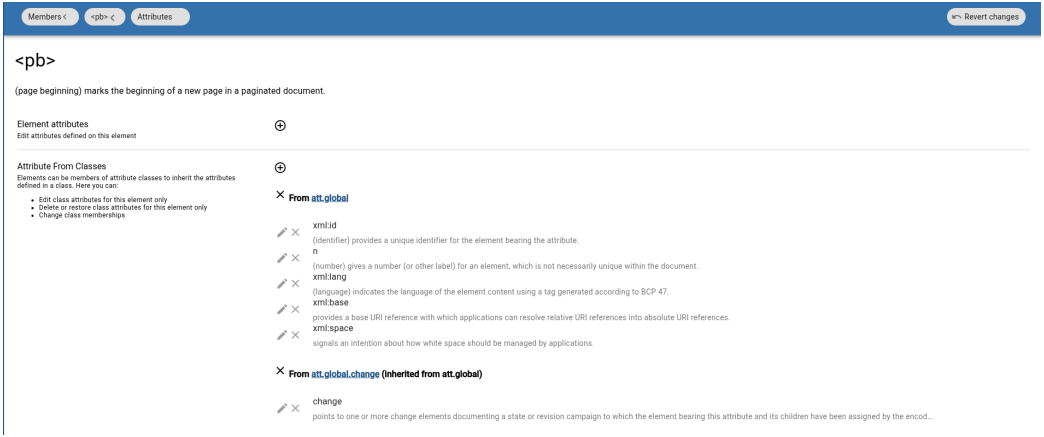


Figure: Find the attribute you want to modify

# Producing ODD's and schemas with ROMA

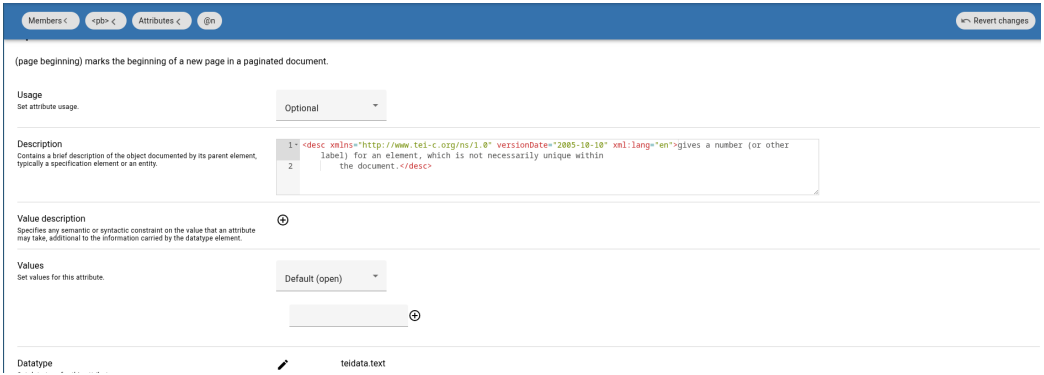


Figure: Change its usage ! It's (almost) automagical !

# The produced ODD

```
    </encodingDesc>
  </teiHeader>
  <text>
    <body>
      <p>TEI for manuscript description</p>
      <schemaSpec ident="tei_ms" start="TEI teiCorpus" prefix="tei_" targetLang="en"
        docLang="en">
        <moduleRef key="header"/>
        <moduleRef key="linking"/>
        <moduleRef key="core"/>
        <moduleRef key="tei"/>
        <moduleRef key="textstructure" except="div1 div2 div3 div4 div5 div6 div7"/>
        <moduleRef key="namesdates"/>
        <moduleRef key="transcr"/>
        <moduleRef key="figures"/>
        <moduleRef key="msdescription"/>
        <elementSpec ident="pb" mode="change">
          <attList>
            <attDef ident="n" mode="change" usage="req"/>
          </attList>
        </elementSpec>
      </schemaSpec>
    </body>
  </text>
</TEI>
```

I find this specification too broad. I would remove some modules. Do you want to describe figures? Are you interested in named entities?

# Producing ODD's and schemas with ROMA: exporting the ODD

- Multiple output formats.
- The most important is ODD for scientific purposes.
- You'll need RNC/RNG for validation purposes.
- You won't be able to retrieve your ODD from an RNG or any other schema file.
- Oxygen can also produce the schema from the ODD's.

# Making a reference to the schema in your TEI file

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE TEI SYSTEM "/home/mgl/Bureau/These/Edition/hyperregimiento-de-los-principes/Dedans/XML/schemas/Esc_Q.dtd">
<?xml-model href="../../schemas/out/ODD_edition.rng" type="application/xml" schematypens="http://relaxng.org/ns/structure/1.0"?>
<TEI xmlns="http://www.tei-c.org/ns/1.0" xml:id="Esc_Q" type="transcription">
 <teiHeader>
```

Figure: Before the root node, the reference to a DTD (in blue, second line) and to my ODD derived schema (in violet, third line).

# A quick look at our ODD

Let's sum up. You can:

- Completely remove elements (they won't be allowed anymore)
- Partially remove elements (they will be allowed in specific contexts only)
- Constrain attribute types, or attribute values; make some attributes mandatory

How does it work ?

# Activate a module and include selected elements

```
<moduleRef key="textcrit" include="listWit witness witEnd witStart app rdg"/>
```

- The `include` attribute means a rebuilding of the module from scratch: absent elements won't be allowed in your encoding
- Note here the choice not to include the `lemma` element, derived from a specific editorial methodology ("variance", etc).

# Make attributes mandatory

```
<elementSpec ident="pb" mode="change">
  <attList>
     <attDef ident="n" mode="change" usage="req"/>
  </attList>
</elementSpec>
```

# Remove attributes at element level

```
<elementSpec ident="material" mode="change" module="msdecription">
    <attList>
        <attDef ident="function" mode="delete"/>
        <attDef ident="target" mode="delete"/>
        <attDef ident="resp" mode="delete"/>
        <attDef ident="ref" mode="delete"/>
    </attList>
</elementSpec>
```

# Constrain attribute values

```
<elementSpec ident="material" mode="change" module="msdecription">
    <attList>
        <attDef ident="type" mode="replace" usage="req">
            <valList mode="add" type="closed">
                <valItem ident="parchment">
                    <desc>The copy medium is parchment.</desc>
                </valItem>
                <valItem ident="paper">
                    <desc>The copy medium is paper.</desc>
                </valItem>
                <valItem ident="mixt">
                    <desc>The copy medium alternates between paper and
                    ↪  parchment.</desc>
                </valItem>
            </valList>
        </attDef>
    </attList>
</elementSpec>
```

# The other side of ODD's: natural language documentation

- Roma is just the first step to create the machine-readable version of your ODD
- As said before, you'll need to create and write the documentation *in natural language*, using the TEI. Example: here.
- You can use any XML editor to do so.

```
<p>Enfin, il est important pour une bonne intelligence du texte, d'aller marquer des divisions structurelles et
  logiques qui se trouvent au sein des paragraphes. C'est pourquoi je propose d'utiliser des balises vides
    <gi>milestone</gi> qui vont marquer, dans un même paragraphe, des énumérations par exemple (méthode moins
  lourde qu'en utilisant <gi>list</gi>): <egXML xmlns="http://www.tei-c.org/ns/Examples">E <w lemma="este"
      pos="PD0MS000">este</w> es el primero capitulo <w lemma="de+este">desta</w>
    <w pos="NCFS000">parte</w> postrimera que <w lemma="estar" pos="VMIP3S0">esta</w> en tres conclusiones.
      <milestone type="structuration_discours" unit="item_rang_1" n="1"/>La primera es que cosa es <w
      lemma="caballería">caualleria</w>. <milestone type="structuration_discours" unit="item_rang_1" n="2"
    />La <w lemma="2" pos="AO0FS0">segunda</w> por que fue fecha e establecida.</egXML></p>
```

Figure: Documenting the use of `milestone` in the edition of the *Regimiento de los prínçipes*.

# An example of complete ODD

- An ODD is an XML-TEI document divided in two parts: the documentation and the schema.
- For structuring, use `div1`...`div9` elements
- You can encode your documentation as you would encode any document
- And put you schema specification (the `schemaSpec`) in a `div1` for instance.

# An example of complete ODD

```
<head>Marques de lecture</head>
<p>l'indique les marques de lecture de deux façons distinctes: <list>
    <item>La façon la plus simple est d'utiliser l'élément <gi>hi</gi> avec plusieurs valeurs d'attribut
        <att>rend</att> (Ajouter un exemple ici)</item>
    <item>Pour éviter les cas d'overlapping, en marquant des marques de lecture de différent niveau,
      j'utilise les balises auto-fermantes <gi>anchor</gi>, qui me permettent de marquer le début et la
      fin d'un passage. Ces balises sont typées (<att>type</att>) et contiennent un sous-type permettant
      de distinguer leur style (sous la ligne, en marge), ainsi qu'un attribut (<att>next</att> ou
        <att>previous</att>) pour indiquer si elles indiquent le début ou la fin de la marque de
      lecture.</item>
  </list><egXML xmlns="http://www.tei-c.org/ns/Examples">sinon <anchor next="#a2SAcLX9bbVCkXl"
      xml:id="a3dutf7XDj8d" type="marque_lecture" subtype="souligne"/>agua salada, aquella <w lemma="agua"
      pos="NCFS000">agua</w> pueden fazer dulçe colándola por la çera, faziendo baçines de çera, et<lb
      break="yes"/><add type="commentaire" place="margin" hand="distincte">agua de
      <unclear>mar</unclear></add>foraçándolos con agujas sotiles de muchos <w lemma="forado">forados</w>
    e espesos.<anchor prev="#a3dutf7XDj8d" xml:id="a2SAcLX9bbVCkXl" type="marque_lecture"
      subtype="souligne"/> Ca segunt <w lemma="el" pos="DA0MS0">el</w>
    <w lemma="filósofo" pos="NCMS000">filósofo</w></egXML></p>
  </div4>
</div3>
<div3>
  <head>Éléments utilisés dans le cadre de la citation du texte</head>
  <p>La citation de fragments du texte se fait par copie et traitement de nœuds XML entre éléments
    <gi>anchor</gi>. Ces éléments sont de type <val>citation</val>. Des <gi>anchor</gi> de type
      <val>reference</val> sont utilisés pour citer une ligne précise du texte, et des <gi>anchor</gi> de type
      <val>reference</val> sont utilisés pour le texte de façon plus générale. Ces éléments doivent
    impérativement être identifiés par un <att>xml:id</att>.</p>
</div3>
<div3> [5 lines]
</div2>
</div1>
<div1 type="Encodage">
  <head>Manuel d'encodage</head>
  <schemaSpec ident="oddbyexample" start="TEI teiCorpus" targetLang="fr" docLang="fr">
    <moduleRef key="tei"/>
    <!--Checking module textcrit-->
    <classRef key="att.textCritical"/>
    <classRef key="model.rdgLike"/>
    <classRef key="model.rdgPart"/>
    <moduleRef key="textcrit" include="listWit witness witEnd witStart app rdg"/>
```

Figure: Documenting the use of `milestone` in the edition of the *Regimiento de los prínçipes*.

# Fine-tune the validation with Schematron

```
<!--The following element can be a child of any elementSpec node-->
<constraintSpec ident="checks_pb"
          xmlns:sch="http://purl.oclc.org/dsdl/schematron"
          xmlns:sqf="http://www.schematron-quickfix.com/validator/process"
          xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
          scheme="isoschematron">
          <constraint>
            <sch:rule context="tei:pb">
              <sch:assert test="matches(@n, '\d+[vr]')">Something is wrong with
              ↪  the foliation of
                this page. Have you forgotten the page side indication
                ?</sch:assert>
            </sch:rule>
          </constraint>
        </constraintSpec>
```

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# Conclusions

- Schemas even minimal, are essential for any scientific publishing project based on TEI.
- TEI-based editions without the publication of XML sources is quite unuseful and won't last much
- Please provide access to your data (and its documentation) !

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# Exercises

- The exercise instructions are in the xml file `bentham.xml`

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# ODDbyexample

- ODDByExample is a tool to produce an ODD using an already existing XML-TEI document
- It will create the ODD that matches this document
- Very useful tool ...
- ... that can be tricky if the XML-TEI example shows inconsistency: they will be embedded into your specification
- Make sure to use this tool with small, carefully checked fragments of your document.
- Link: https://github.com/TEIC/Stylesheets/blob/dev/tools/oddbyexample.xsl

# References


Burnard, Lou. “What Is TEI Conformance, and Why Should You Care?” In: *Journal of the Text Encoding Initiative* 12 (2019). ISSN: 2162-5603.

Marín Sánchez, Ana María. “La Versión Interpolada de Los Castigos de Sancho IV: Edición y Estudio”. PhD thesis. Universidad de Zaragoza, 2003.